

Specification: LLM Response Generation

1 Objective

Generate responses from LLMs for each combination of persona, problem, and model version (small/large).

2 Inputs

File	Format	Description
<code>prompts/personas.jsonl</code>	JSONL	Persona definitions (id, system_prompt)
<code>data/100_problems.csv</code>	CSV	Columns: problem_id, problem

3 Output Format

Generate a CSV file with the following columns:

Column	Type	Description	Example
<code>custom_id</code>	string	<code>{model_name}-{condition}_prob{problem_id}</code>	<code>gpt-4o-mini.Culture_5_prob42</code>
<code>condition</code>	string	Persona ID or Baseline	<code>Culture_5, Small_Baseline, Large_Baseline</code>
<code>problem_id</code>	integer	0-99	<code>42</code>
<code>model</code>	string	Model name used	<code>gpt-4o-mini</code>
<code>content</code>	string	Generated response text	<code>The tangent line is $y = 10x - 16$...</code>
<code>problem_text</code>	string	Original problem text	<code>Find the equation...</code>

4 Conditions to Generate

Condition	Model Version	Persona	System Prompt
<code>{persona.id}</code>	Small	Yes	From <code>personas.jsonl</code>
<code>Small_Baseline</code>	Small	No	None (send only problem)
<code>Large_Baseline</code>	Large	No	None (send only problem)

5 Generation Parameters

6 Example API Call (OpenAI)

```
response = client.chat.completions.create(  
    model="gpt-4o-mini",  
    messages=[  
        {"role": "system", "content": system_prompt}, # For personas only  
        {"role": "user", "content": problem_text}  
    ],  
)
```

Parameter	Value	Reason
temperature	0.7	Balances creativity and coherence
max_tokens	16000	Accommodates detailed responses
top_p	0.95	Standard for diverse outputs

```
temperature=0.7,
max_tokens=16000
)
```

7 Notes for Users

- Use your preferred LLM provider (OpenAI, Anthropic, Together.ai, Google, etc.)
- Adapt the API call to your provider's format
- Implement rate limiting and exponential backoff for retries
- Save responses incrementally after each problem to avoid data loss
- The `Small.Baseline` and `Large.Baseline` conditions use **no system prompt** – only the problem text